

Time Series Prediction

A comparative study of **5 forecasting models** - from classical ARIMA to deep state-space networks - across 5 univariate & multivariate datasets and three horizons

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01

Problem & Motivation

Motivation

Almost every operational decision is a bet on a **future value of a measured signal**. Better forecasts mean better decisions

ENERGY & ENVIRONMENT

Plan supply, warn early

Appliance load and air-quality (PM, NO₂, O₃) forecasts drive grid scheduling and public-health alerts

FINANCE

Price under uncertainty

Asset prices such as Bitcoin are noisy and non-stationary - a stress test for any forecaster

NATURAL & SCIENTIFIC CYCLES

Anticipate the rhythm

Phenomena like the sunspot cycle are strongly periodic yet hard to pin down precisely

THE QUESTION

No single model wins everywhere. Which family - statistical, tree-based, or deep - earns its complexity, and when?

Problem Definition

Given a multivariate series of **C** channels observed over time, we use a fixed **look-back window** of length **L** to predict the next **horizon** of **H** steps for a chosen set of target channels

Window → horizon

$$\hat{\mathbf{Y}}_{t+1:t+H} = f_{\theta}(\mathbf{X}_{t-L+1:t})$$

UNIVARIATE

One channel predicts itself:

$$C = M = 1$$



MULTIVARIATE

Many channels, several targets:

$$C > 1, M \geq 1$$



where $\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{L \times C}$ is the input window, $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times M}$ the prediction over M target channels, and f_{θ} the learned model. We minimise mean-squared error over the training windows:

Objective

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \left\| \mathbf{Y}^{(i)} - f_{\theta}(\mathbf{X}^{(i)}) \right\|_2^2$$

OUR SETTING

Look-back $L = 48$, Horizons $H \in \{12, 24, 48\}$

Scope

We hold the data pipeline, look-back, horizons and metrics **fixed** , and vary only the model.
This isolates the effect of modelling choice across a deliberately diverse set of signals

5 MODELS - 3 FAMILIES

- ARIMA - statistical
- XGBoost - trees
- iTransformer - deep
- TimeMixer - deep
- TSMamba - deep

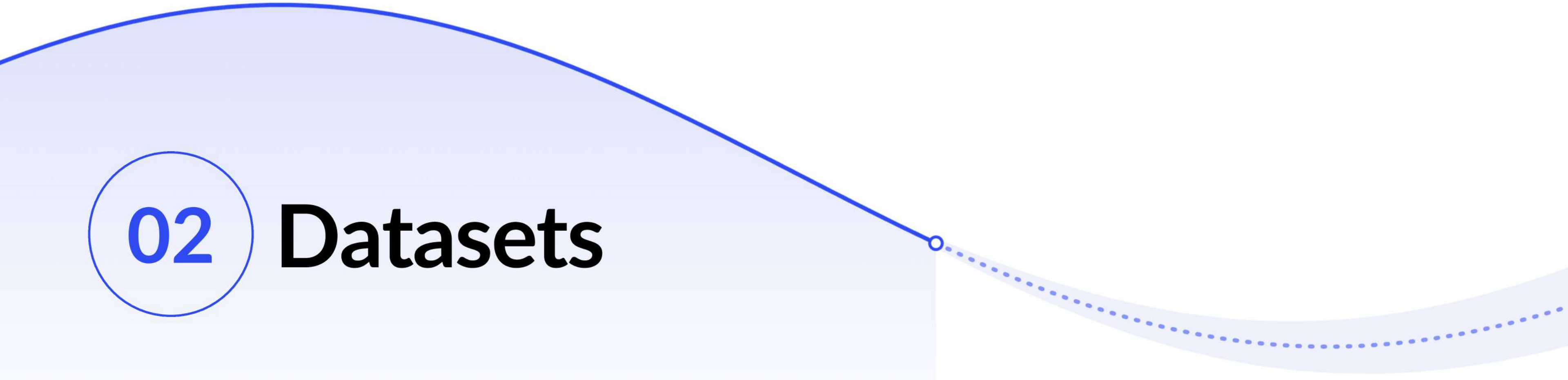
5 DATASETS - UNI + MULTI

Identical windowing, train/val/test split,
and scaling per dataset

Same evaluation metrics on the original
scale

Identical windowing, train/val/test split,
and scaling per dataset

3 HORIZONS - 12 / 24 / 48

A decorative blue wave graphic that starts on the left, peaks, and then descends into a dotted line on the right. The area under the solid part of the wave is filled with a light blue gradient.

02 Datasets

Data Overview

Two univariate series and three multivariate sets - spanning natural cycles, energy, urban air quality and finance - give a deliberately broad test bed

DATASET	DOMAIN	TYPE	TARGETS	FEATURES	SAMPLING
Sunspots	Solar activity	Univariate	1	3	Monthly
Appliances Energy	Building energy	Multivariate	2	28	10-min
Beijing Air Quality	Urban air	Multivariate	6	15	Hourly
Hanoi Air Quality	Urban air	Multivariate	7	18	Hourly
Bitcoin	Finance	Univariate	1	1	Hourly

Targets range from **1 to 7 channels** and dynamics from smooth periodic (sunspots) to sharp and noisy (Bitcoin) - stressing each model differently

Preprocessing Pipeline

01 Parse & sort time

Build a single datetime index

02 Clean missing values

Forward-fill, then fill residual gaps with the training-set median

03 Add cyclical features

Append sin/cos encodings of hour and month to expose periodic structure

04 Scale

Standardise features & targets with statistics fit on the training split only

Standardization

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

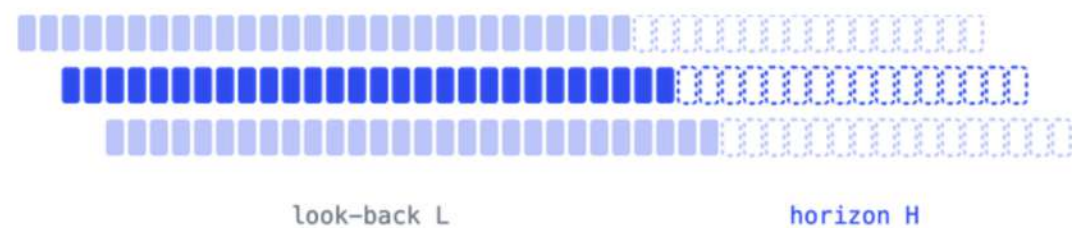
Fitting scalers on the training split only prevents **look-ahead leakage** from validation/test into the model's input statistics. Predictions are inverse-transformed back to the original scale before scoring

The same feature set, cleaning rules and scaler are reused for every model - so differences in results reflect the model, not the data preprocess

Windowing & Splits

Each training example is a contiguous look-back window of L steps paired with the following H steps. The window slides one step at a time across the series

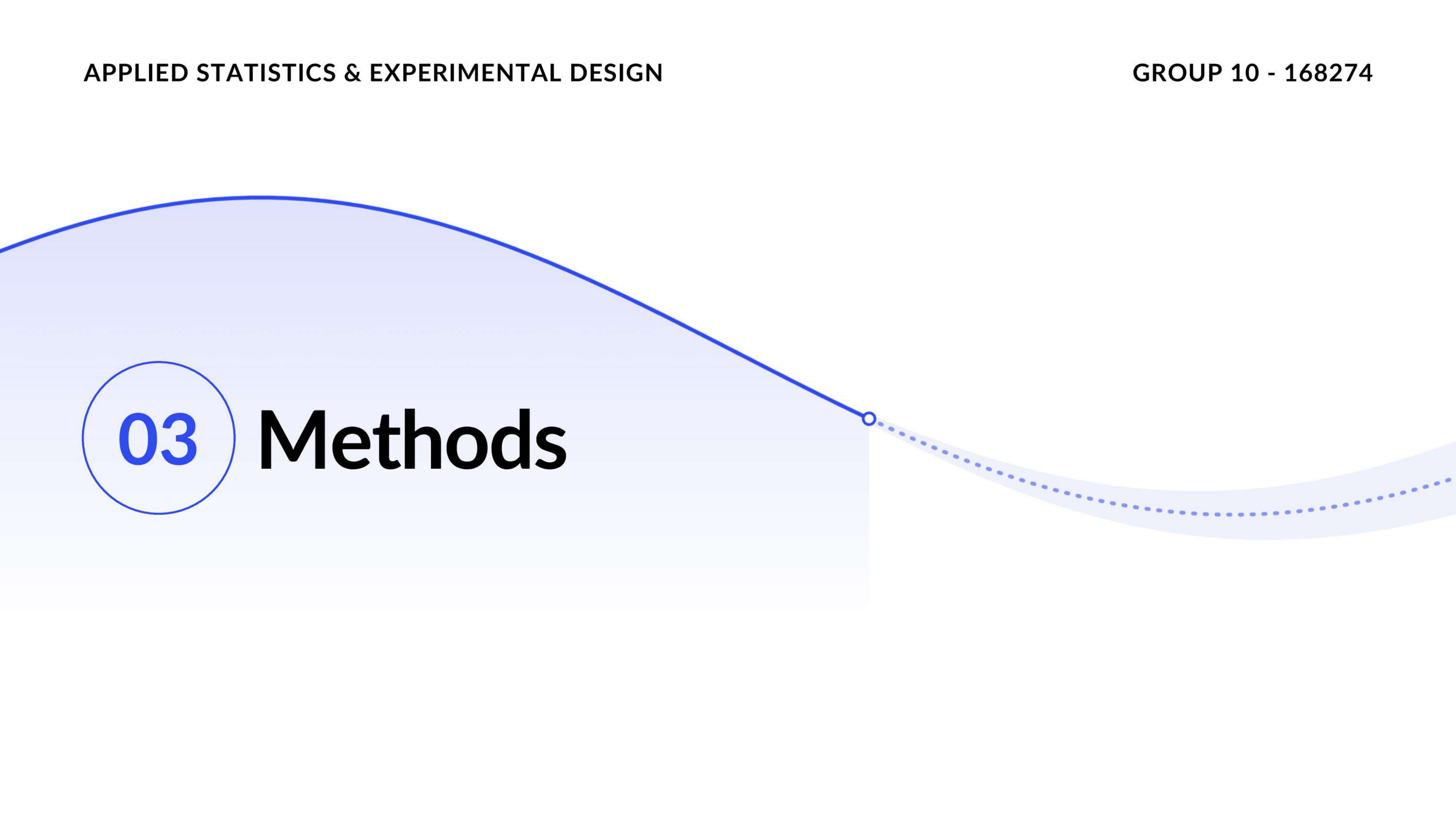
Example i $(\mathbf{X}_{i:i+L}, \mathbf{Y}_{i+L:i+L+H})$



CHRONOLOGICAL SPLIT



Splits are taken **in time order** so the test set is strictly the future. Window boundaries overlap by L so no target is lost at the seams



03 Methods

Methods

STATISTICAL

ARIMA

Linear model of a series' own past. Few parameters, strong assumptions, fully interpretable

GRADIENT BOOSTING

XGBoost

Ensemble of regression trees on a flattened window. Non-linear, no temporal prior

DEEP LEARNING

iTransformer • TimeMixer • TSMamba

Attention, multiscale mixing, and selective state spaces - learn temporal & cross-channel structure

ARIMA

ARIMA assumes a **stationary** series - constant mean and autocovariance. Non-stationary series are made stationary by **differencing** d times

Differencing $\nabla^d y_t = (1 - B)^d y_t, \quad \nabla y_t = y_t - y_{t-1}$

On the differenced series $y'_t = \nabla^d y_t$, ARIMA combines an autoregressive term (past values) with a moving-average term (past errors)

ARIMA (p, d, q)
$$y'_t = \underbrace{\sum_{i=1}^p \phi_i y'_{t-i}}_{AR(p)} + \varepsilon_t + \underbrace{\sum_{j=1}^q \theta_j \varepsilon_{t-j}}_{MA(q)}$$

OUR CONFIGURATION

3

AR ORDER - P

0

DIFF. - D

3

MA ORDER - Q

IMPLEMENTATION NOTE

Both AR and MA terms are estimated by ordinary least squares - a lightweight, from-scratch approximation rather than full maximum-likelihood. Forecasts are produced by one-step-ahead rolling prediction.

XGBoost

The forecast is an additive ensemble of regression trees - each new tree corrects the errors of the current model with a shrinkage rate

Additive model

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(\mathbf{x}_i)$$

Each tree is fit to minimise a regularised objective trading fit against complexity:

Objective

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i^{(t)}) + \Omega(f_t)$$

$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|\mathbf{w}\|_2^2$ penalises the number of leaves T and leaf weights w

OUR CONFIGURATION

200

TREES

0.05

LEARNING RATE

6

MAX DEPTH

0.8

SUBSAMPLE

0.8

COL-SAMPLE

IMPLEMENTATION NOTE

The look-back window is flattened to one feature vector. One model per target predicts the final step of the horizon - trees carry no built-in notion of time order

XGBoost

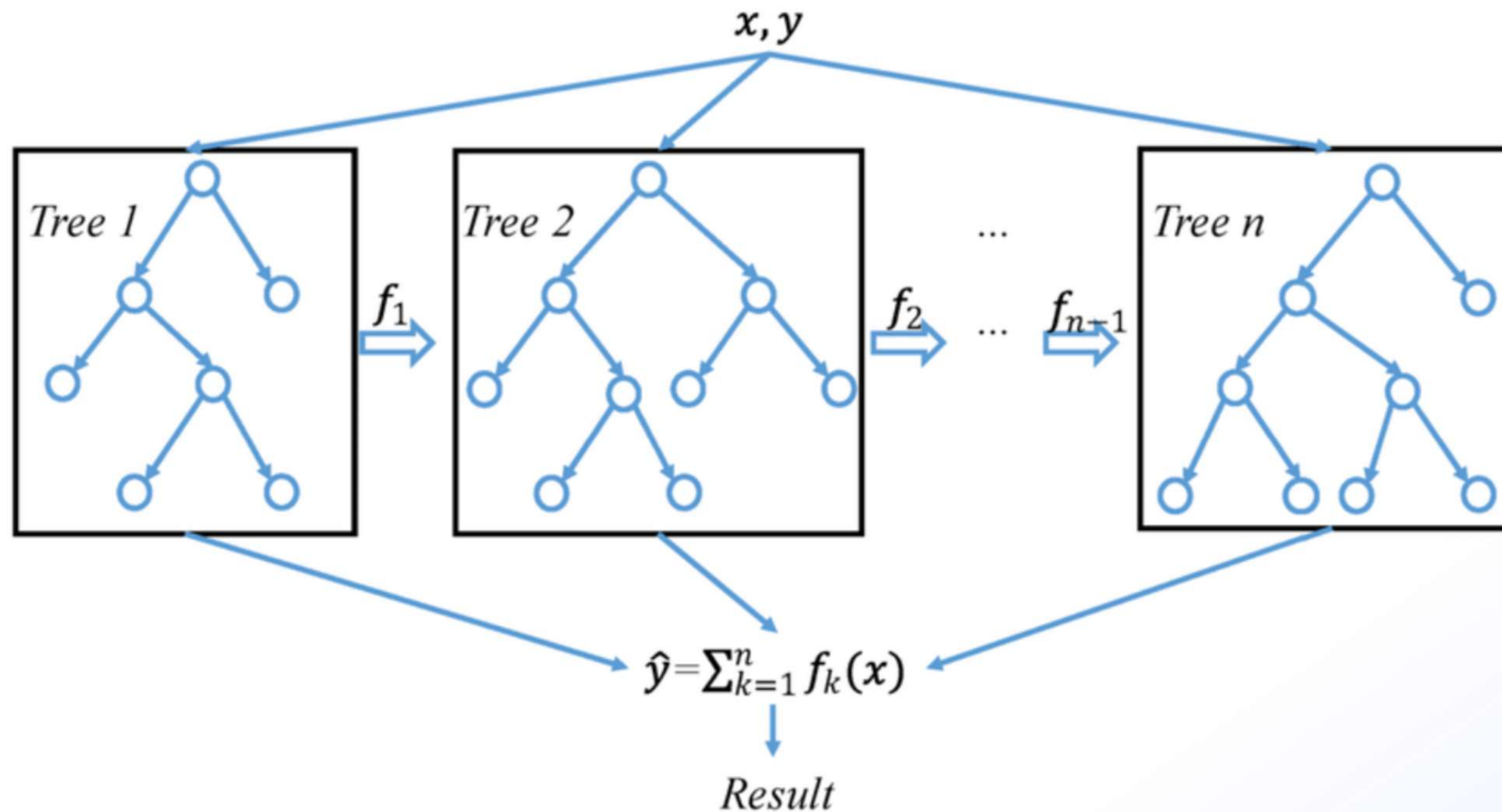


Figure 1. A general architecture of XGBoost. Wang, Y., Pan, Z., Zheng, J., Qian, L., & Li, M. (2019). "A hybrid ensemble method for pulsar candidate classification". *Astrophysics and Space Science*, 364(8), 139.

iTransformer

iTransformer inverts the usual layout: instead of one token per time step, the entire history of each variate is embedded into a single token. Attention then models cross-channel dependencies.

Reversible norm $\tilde{\mathbf{x}} = \frac{\mathbf{x} - \mu}{\sigma} \odot \gamma + \beta$

Variate embedding $\mathbf{h}_n = \text{Embed}(\tilde{\mathbf{X}}_{:,n}) \in \mathbb{R}^d, n = \overline{1, C}$

OUR CONFIGURATION

64

D-MODEL

4

ATTENTION HEADS

3

ATTENTION LAYERS

5e-4

LEARNING RATE

0.2

DROPOUT

QKV in self-attention are the C variate tokens, so attention weights are a learned channel-correlation matrix

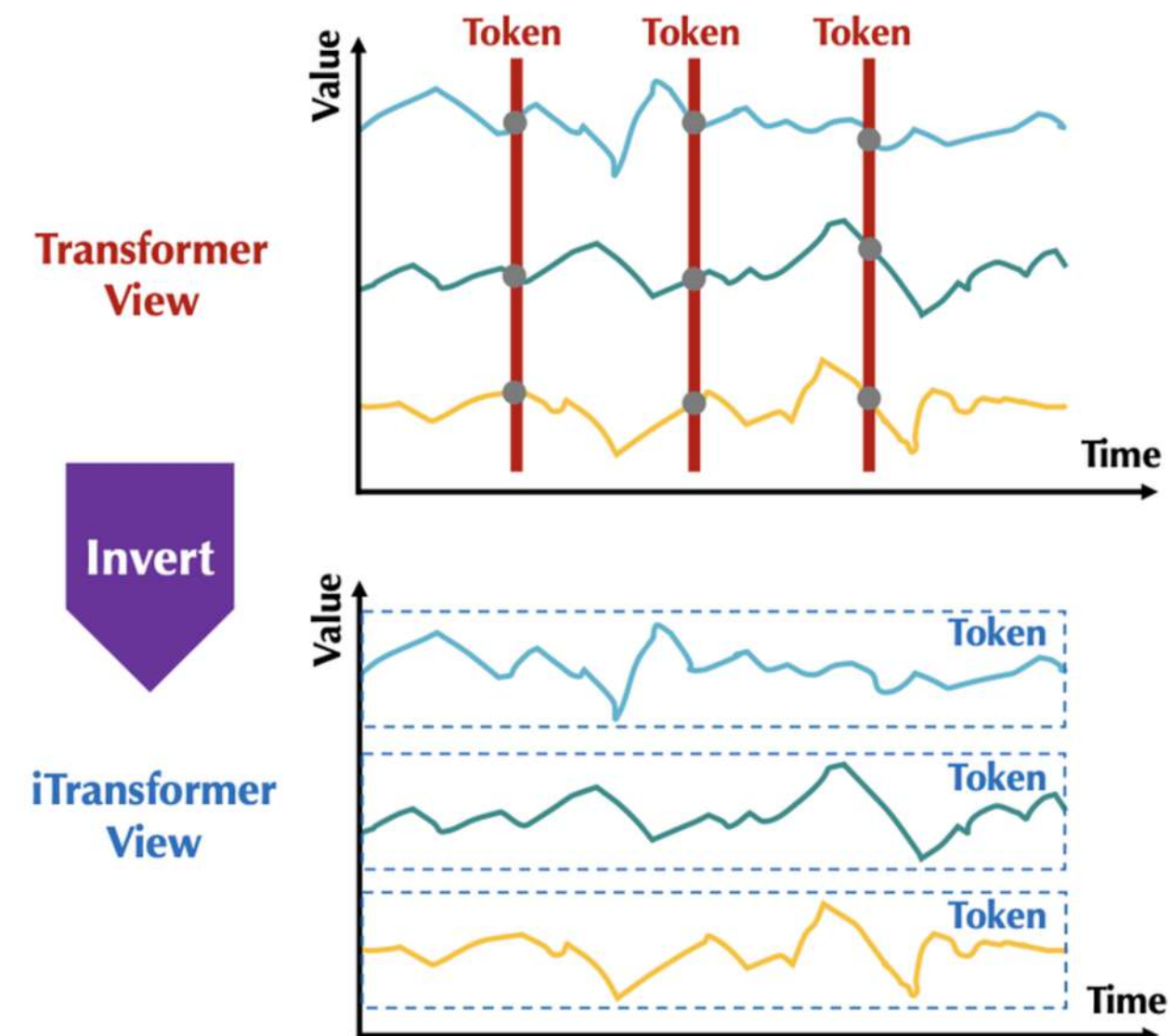


Figure 2. Comparison between the vanilla Transformer and iTransformer. Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, and M. Long (2023). "iTransformer: Inverted Transformers Are Effective for Time Series Forecasting". In International conference on learning representations (Vol. 2024).

TimeMixer

Each window is split into a smooth trend and a fluctuating seasonal part, then viewed at several temporal resolutions via average-pooling

Series decomposition $\tau_t = \text{AvgPool}(\mathbf{x})_t, s_t = x_t - \tau_t$

Multiscale $\mathbf{x}^{(\ell)} = \text{DownSample}(\mathbf{x}^{(\ell-1)}), \ell = \overline{1, M}$

In each mixing block, fine-grained seasonality is mixed bottom-up while coarse trends are mixed top-down:

Mixing $\mathbf{s}^{(\ell+1)} += \text{MLP}_{\uparrow}(\mathbf{s}^{(\ell)}), \boldsymbol{\tau}^{(\ell)} += \text{MLP}_{\downarrow}(\boldsymbol{\tau}^{(\ell+1)})$

Predictions from every scale are projected to the horizon and summed:

$$\hat{\mathbf{Y}} = \sum_{\ell} \text{Proj}_{\ell}(\mathbf{x}^{(\ell)})$$

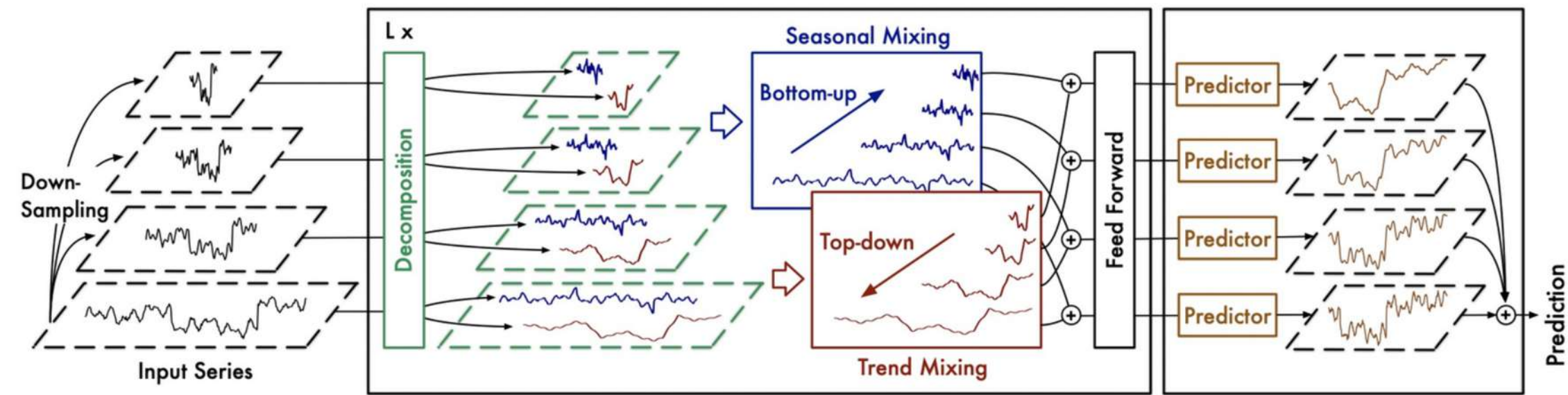


Figure 3. Overall architecture of TimeMixer. Wang, S., Wu, H., Shi, X., Hu, T., Luo, H., Ma, L., ... & Zhou, J. (2024, May). "Timemixer: Decomposable multiscale mixing for time series forecasting". In International conference on learning representations (Vol. 2024).

OUR CONFIGURATION

32	2
D-MODEL	DOWN SAMPLING
5e-4	0.3
LEARNING RATE	DROPOUT

TSMamba

A **state-space model** maps an input sequence through a hidden state with linear dynamics - a continuous system that is then discretised

Continuous SSM $\mathbf{h}'(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t), \mathbf{y}(t) = \mathbf{C}\mathbf{h}(t)$

Discretise $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}), \bar{\mathbf{B}} \approx \Delta\mathbf{B}$

Recurrence $\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}\mathbf{x}_t, \mathbf{y}_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}\mathbf{x}_t$

For forecasting, the window is cut into non-overlapping patches and encoded in both directions - forward and reversed - so each patch sees its full context

Bidirectional encoding $\mathbf{h} = \text{LN}\left(\overrightarrow{\text{Mamba}}(\mathbf{p}) + \overleftarrow{\text{Mamba}}(\mathbf{p})\right)$

OUR CONFIGURATION

32

D-MODEL

8

D-STATE

5e-4

LEARNING RATE

0.3

DROPOUT

TSMamba

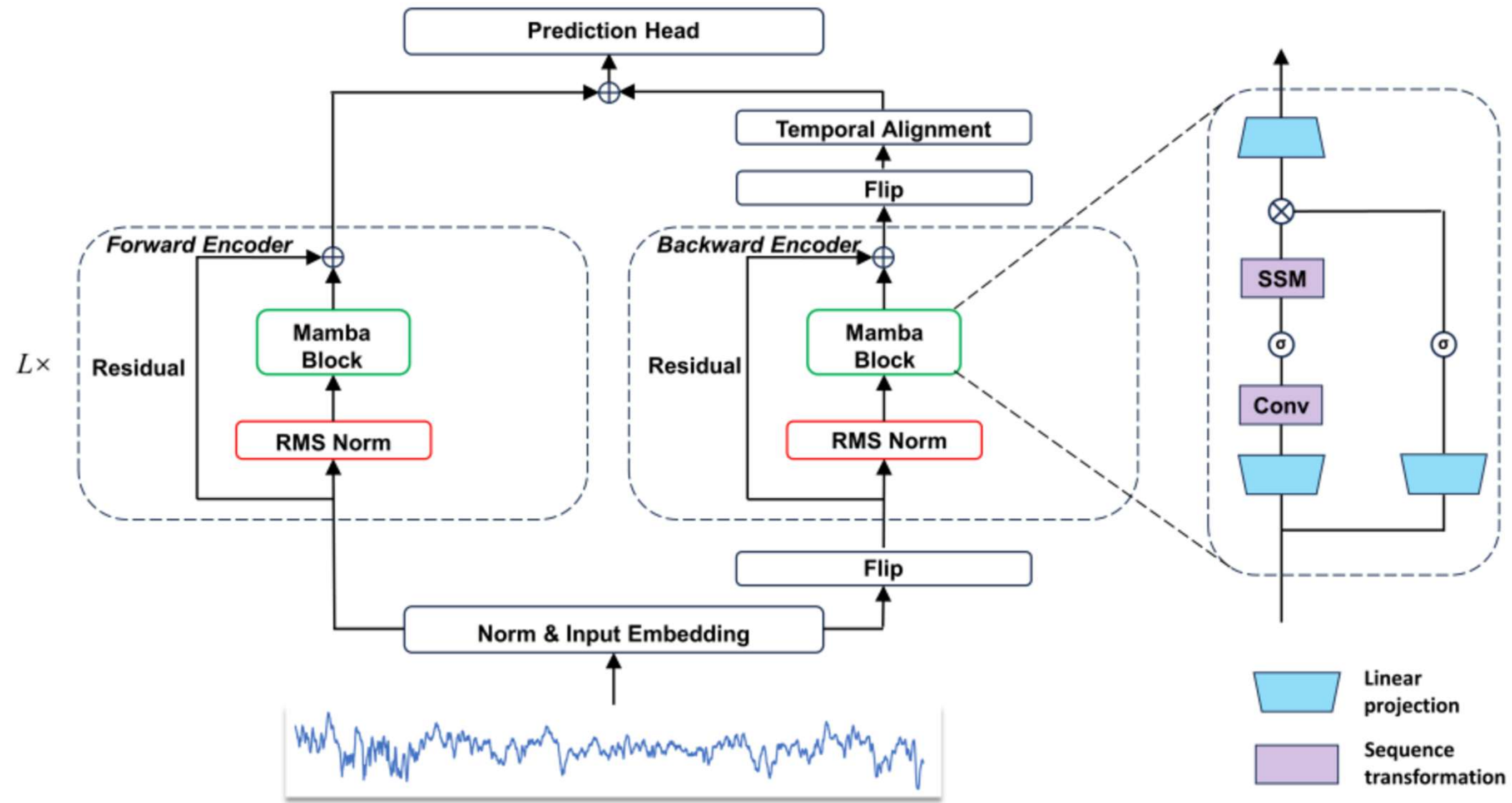
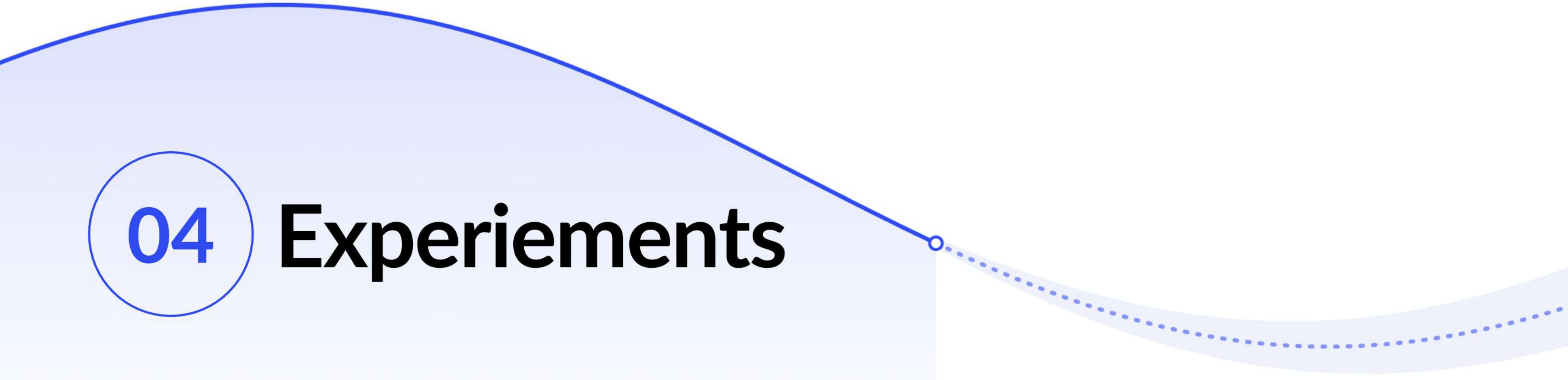


Figure 4. TSMamba Architecture. Ma, H., Chen, Y., Zhao, W., Yang, J., Ji, Y., Xu, X., ... & Yang, G. (2024). "A mamba foundation model for time series forecasting"

A decorative blue wave graphic that starts on the left, peaks, and then descends into a dotted line on the right.

04 Experiements

EXPERIMENTAL SETTINGS

- Look-back $L = 48$
- Horizons $H \in \{12, 24, 48\}$
- Loss: MSE (on standardised targets)
- Number of epochs: up to 200 epochs (early stopping 20)
- Best checkpoint chosen by validation loss
- Metrics computed on the original scale after inverse-transform

5 models $\times$ 5 datasets $\times$ 3 horizons = 75 evaluation runs

EVALUATION METRICS

Four complementary metrics - two absolute-error, one scale-free, one variance-explained - give a rounded picture of accuracy

MEAN ABSOLUTE ERROR

Average error magnitude, in the target's own units

$$\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$$

ROOT MEAN SQUARED ERROR

Penalises large errors more - sensitive to spikes

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}$$

WEIGHTED MAPE

Scale-free, comparable across datasets of different magnitudes

$$\text{WMAPE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i|}$$

COEFFICIENT OF DETERMINATION

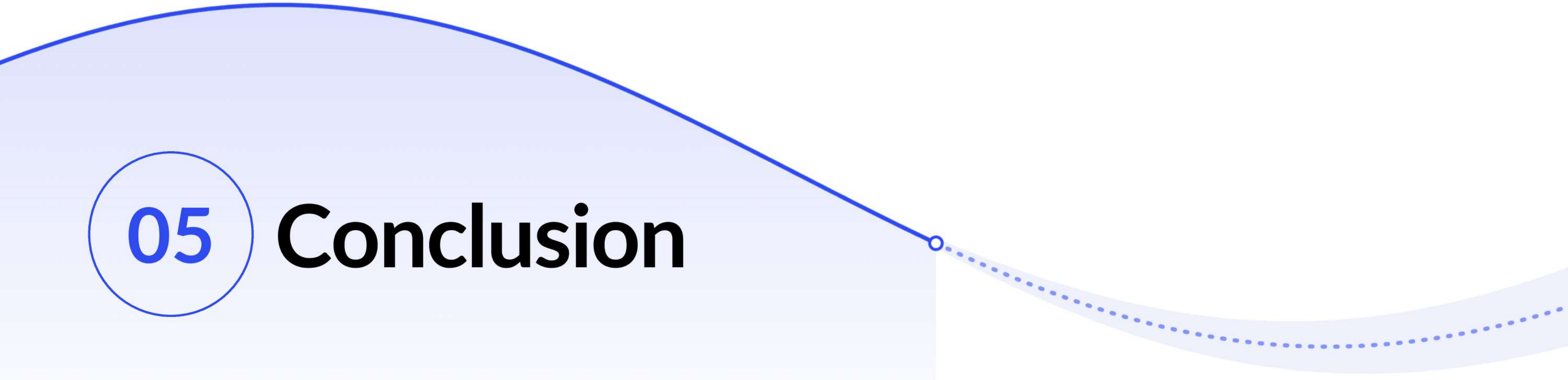
Fraction of variance explained. 1 is perfect, 0 is the mean baseline.

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

RESULTS & DEMO



<https://huggingface.co/spaces/nvtkienn/time-series-prediction-demo>

A decorative blue wave graphic that starts as a solid line on the left, curves down to a small circle, and then continues as a dotted line towards the right. The area under the solid part of the wave is filled with a light blue color.

05 Conclusion

SUMMARY

- Built a **reusable & scalable** pipeline with shared preprocessing, windowing & metrics over 5 datasets
- Benchmarked **5 models across 3 families and 3 horizons**
- Implemented all models **from scratch**; trained three deep models end-to-end

LIMITATIONS

- Deep models are **not exhaustively tuned**
- From-scratch ARIMA & XGBoost are correct but **slower** than optimised libraries
- **Short look-back** (L=48)



Improve in future works

A decorative background featuring a solid blue curve that starts on the left, peaks, and then descends towards the center. A dotted blue line continues from the end of the solid curve, extending towards the right side of the slide. The area under the solid curve is filled with a light blue color, and the area under the dotted line is also filled with a light blue color.

Thank you for your attention!